Advanced Derivatives

FIN-503

Assignment 3

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Exercise 2

The dynamic SABR model has model specifications:

$$\begin{aligned} dF_t &= \sigma_t F_t^{\beta} dW_t^F \\ d\sigma_t &= \alpha \sigma_t dW_t^{\sigma} \\ dW_t^F dW_t^{\sigma} &= \rho dt \end{aligned}$$

with $0 \leq \beta \leq 1$ and $\alpha \geq 0$. The parameters ρ and α are assumed to be time dependent:

$$\begin{aligned} \rho(t) &= \rho_0 e^{-at} \\ \alpha(t) &= \alpha_0 e^{-bt} \end{aligned}$$

with a and b two constants, parameters of the model. With such model, the volatility has closed form (see: https://arxiv.org/pdf/2407.20713):

$$\sigma^{model}(K, \tau; F) = \frac{1}{\omega}\left(1 + A_1(\tau)\log\frac{K}{F} + A_2(\tau)\left(\log\frac{K}{F}\right)^2 + B(\tau)\tau\right)$$

with

$$\omega = \frac{F^{1-\beta}}{\sigma_0}$$

The functions A_1, A_2 and B are given by:

$$A_1(\tau) = \frac{\beta - 1}{2} + \frac{\eta_1(\tau)\omega}{2}$$

$$A_2(\tau) = \frac{(1-\beta)^2}{12} + \frac{1 - \beta - \eta_1(\tau)\omega}{4} + \frac{4\nu_1(\tau) + 3(\eta_2(\tau) - 3\eta_1^2(\tau))}{24}\omega^2$$

$$B(\tau) = \frac{1}{\omega^2}\left(\frac{(1-\beta)^2}{24} + \frac{\beta\eta_1(\tau)\omega}{4} + \frac{2\nu_2(\tau) - 3\eta_2(\tau)}{24}\omega^2\right)$$

with η_1, η_2, ν_1 and ν_2 given by:

$$\nu_1(\tau) = \frac{6\alpha_0^2}{(2b\tau)^3}\left[\frac{(2b\tau)^2}{2} - 2b\tau + 1 - e^{-2b\tau}\right]$$

$$\nu_2(\tau) = \frac{6\alpha_0^2}{(2b\tau)^3}\left[2(e^{-2b\tau} - 1) + 2b\tau(e^{-2b\tau} + 1)\right]$$

$$\eta_1(\tau) = \frac{2\alpha_0\rho_0}{\tau^2(a+b)^2}\left[e^{-(a+b)\tau} - (1 - (a+b)\tau)\right]$$

$$\eta_2(\tau) = \frac{3\alpha_0^2\rho_0^2}{\tau^4(a+b)^4}\left[e^{-2(a+b)\tau} - 8e^{-(a+b)\tau} + (7 + 2(a+b)\tau(-3 + (a+b)\tau))\right]$$

Our goal is to find the parameters $\sigma_0, \alpha_0, \beta, \rho_0, a$ and b such that this model matches the market data given in the following table.

K(% of price)	$\sigma^{mkt}(\%)$ for $T = 0.25$	$\sigma^{mkt}(\%)$ for $T = 0.5$	$\sigma^{mkt}(\%)$ for $T = 1$	$\sigma^{mkt}(\%)$ for $T = 2$
0.88	32.21	32.07	30.11	28.16
1	29.79	29.63	28.22	26.54
1.12	27.52	27.42	26.34	24.95

In order to do so, we define our objective function as:

$$\min_{\sigma_0,\alpha_0,\beta,\rho_0,a,b} \sum \left(\sigma^{model}(K,\tau;F) - \sigma^{mkt}(K,\tau;F)\right)^2$$

Note that K is expressed as a percentage of the current price F, and thus the terms $\frac{K}{F}$ can directly be replaced by the values in the table. To get the optimal parameters, we use `least_squares` from `scipy.optimize`. Also, when minimizing, we choose the following initial parameters:

- $\sigma_0 = 0.2979$: it represents the initial volatility of the model, so we take the market volatility of an ATM option with smallest maturity.
- $\alpha_0 = 0.5$: It is defined as a positive constant, the magnitude is arbitrary chosen
- $\beta = 0.8$: It is bounded between 0 and 1, when running a first simulation, we obtained a large value. We decided to fix it closer to one to optimize the search.
- $\rho_0 = -0.3$: It is known that this coefficient is negative from the implied volatility smile. Magnitude is chosen arbitrary.
- $a = b = 0.1$: Arbitrary chosen

Using our solver, we obtain the following list of calibrated parameters:

$$\hat{\sigma}_0 = 0.2944, \quad \hat{\alpha}_0 = 0.4059, \quad \hat{\beta} = 1$$

$$\hat{\rho}_0 = -1 \quad \hat{a} = 0, \quad \hat{b} = 0.09359$$

and the final MSE is given by `5.669e-05`. We have made the assumption that the factor w does not depend on the maturity of the option. In reality, F is the value of a forwrd, and thus depends on the maturity. Here we suppose that it is a constant. What we can do is set $F = 1$, and let our model optimize for σ_0, that in reality is a scaled value by a factor $F^{1-\beta}$. This does not affect the performance and fit of our model, but gives a scaled value of σ_0.

Using those parameters, we can put them in our equation for σ^{model} to compute the cross-sectional implied vol smiles as a function of strike, for each expiration in the table. We also plot the market data points to see our model performance. An analysis of the cross-sectional plot (Figure.1) reveals clear systematic errors in the calibration fit. The model demonstrates a consistent bias across the term structure, as it underestimates market volatilities for short-dated expirations ($T = 0.25$ and $T = 0.5$) and overestimates them for long-dated expirations ($T = 1.0$ and $T = 2.0$). This error pattern is coupled with a systematic skew mispricing that "twists" over time: the model is most accurate for low strikes at short maturities but becomes progressively more accurate for high strikes as maturity increases. This structural mismatch suggests the model is unable to capture both the market's term structure level and the term structure of its skew.

Comments on some estimator values and their signification:

1. $\hat{\beta} = 1$: For this value of β, there exists a closed form solution to the problem. Indeed, we get $dF_t = \sigma_t F_t dW_t^F$, the standard log-normal(Black-Scholes) dynamic. This implies that the market data is best described by a model where volatility is proportional to the forward price.
2. $\hat{a} = 0$ and $\hat{\rho}_0 = -1$: It means that ρ does not depend on time. The correlation between the two brownian motion is -1, we can thus express one as a linear transformation of the other. There's, in reality, only one source of volatility in the system.

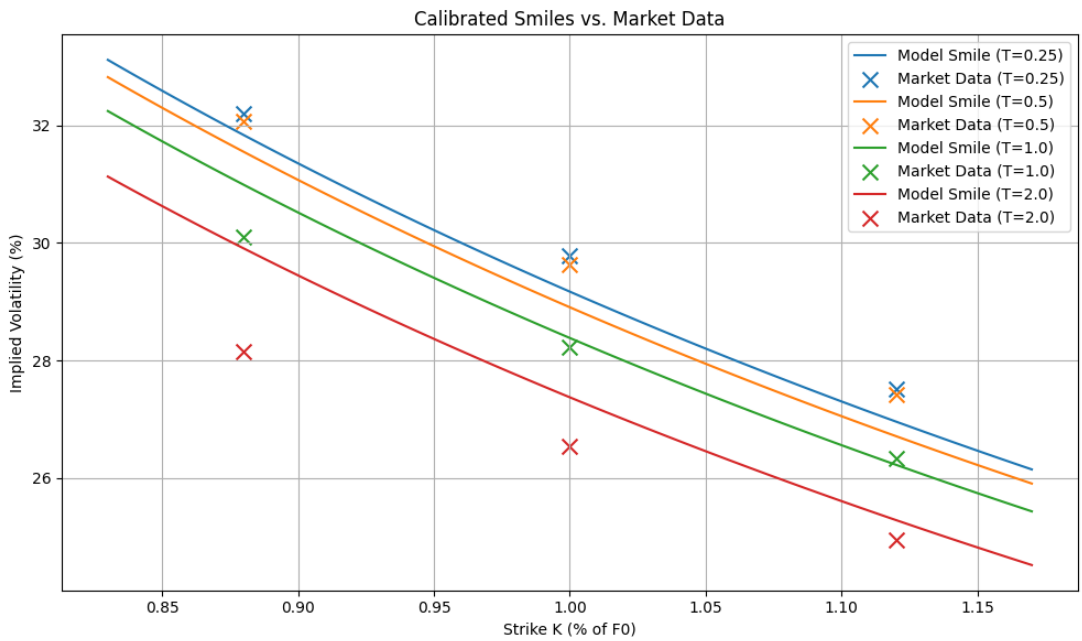

Figure 1: Cross-sectional implied volatility smiles

We have also decided to show the surface in the following plot[1] :

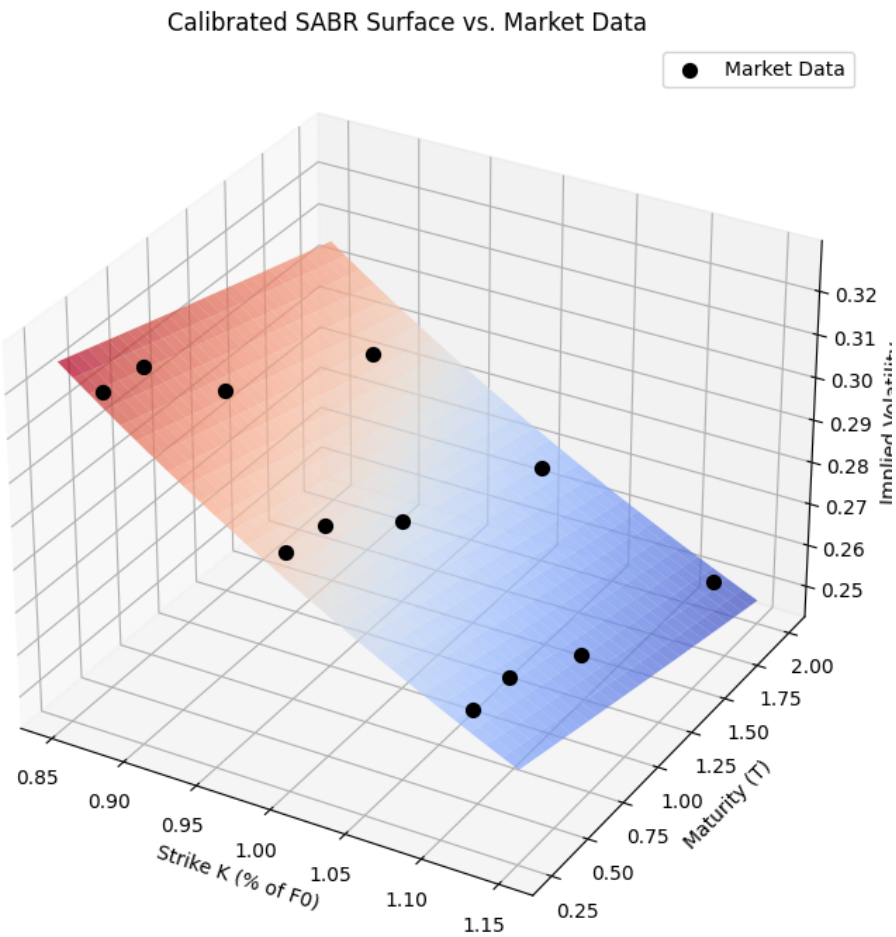

Figure 2: Surface

[1]ChatGPT has helped us make clean plots for both Figures 1 and 2.